

A neuron’s tuning is fixed by its correlations with other neurons

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Abstract

What a neuron represents is usually established by relating its activity to a stimulus. We ask whether the same facts are contained in the correlation matrix of a recorded population, which keeps the pairwise relations among the neurons and discards the stimulus. A transformer that sees only the correlation matrix of 512 sampled neurons, with no labels and no reference population in its input, decodes the preferred orientation of mouse visual cortex neurons from in-vivo MICrONS responses to a mean error of 25° (uninformed: 45° ; best constant guess: 41° ; 28° with every orientation weighted equally) and their receptive-field position at $R^2 = 0.23$; on the digital-twin substrate, 20° (24.5° balanced) and 0.48. Asked whether a neuron prefers horizontal or vertical, it is right for 79% of neurons in vivo and 87% on the twin, against 56% for the majority answer; it does not resolve the obliques. The ordering is recoverable because the relations between orientation classes are close to circulant, which fixes the class structure up to rotation and reflection; the absolute frame is recoverable only because of a departure from that symmetry. The departure survives equal representation of every orientation and disjoint training and test stimuli, and it is an anisotropy of the signal correlations: in V1 the neurons preferring the dominant orientation have the most similar responses to one another, and a model population with only that property recovers the frame. Across 33 mice of the Allen Brain Observatory, with relations from drifting gratings, orientation transfers to unseen animals only when a sampled population mixes animals; populations drawn from one animal are decoded only 0.3° to 1.8° below the best constant, a limit of the training diversity one animal provides rather than of its circuit, since single-scan populations in MICrONS reach 9° to 10° below it (31° to 32°). For receptive fields, what crosses animals is where each animal’s imaged patch looks on the screen, not the layout of neurons within it.

1 Introduction

A neuron in visual cortex can be characterised by what it responds to: vertical gratings, or the upper left of the screen. This is conventionally established with the stimulus. We ask whether the same facts are contained in a description that mentions no stimulus. For a population that watched the same movie, one can compute the correlation of every pair of neurons and then discard the movie. Does the resulting matrix determine which neuron prefers which orientation, and where each neuron’s receptive field lies?

The idea that content can be read from relations is old. Shepard and Chipman proposed that a representation need not resemble what it stands for as long as the similarities among representations mirror the similarities among the things represented [1], and representational similarity analysis turned this into a method that compares brains and models through their similarity structures rather than their coordinates [2]. Structuralist accounts of experience claim that what a state represents, or how it feels, is fixed by its position in a space of relations [3]. Lässig [4] formulates the corresponding question for neural networks and gives the decoding framework we use here.

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Two lines of work come close. Tsodyks et al. [5] showed that the spike-triggered average of spontaneous population activity reproduces the map of a neuron’s preferred orientation, with no stimulus present; the tuning was read from that pattern by comparison with the evoked map, a stimulus-labelled reference. Unsupervised manifold methods recover a neuron’s tuning from the structure of population activity alone, without measuring the variable it encodes [6], but the recovered coordinate is defined only up to the symmetries of the latent space, and it is related to the measured variable by aligning to it afterwards. In both cases the relations locate a neuron within the structure of the encoded variable, and something outside the relations, the evoked map or the measured behaviour, says which value that location corresponds to. We ask whether the relations among neurons fix the content of each neuron when the stimulus is discarded and there is no labelled neuron, no map and no behaviour to compare with. We also ask whether that content is readable across thousands of individually identified neurons and across animals. The nearest machine-learning work reads properties of a neuron from population activity: NuCLR learns a neuron’s identity from its population context with a permutation-equivariant transformer and transfers cell type and brain region to unseen animals [7]. It sees the activity itself. Our decoder sees only the correlation matrix, so it answers what the relations alone determine, and its target is stimulus tuning rather than cell class. This relational setting is common: recordings under natural behaviour have no controlled stimulus, and neurons from different sessions or animals share no stimulus-aligned coordinates, but they do share the structure of their correlations.

We answer this question with a decoder whose input is the correlation matrix of a sampled population, one row per neuron, and nothing else, and whose output is, per neuron, either a preferred orientation or a receptive-field position. It is trained on labelled populations and scored on neurons it never saw, at two scales. Within one brain we use the MICrONS release [8, 9]: 12,894 neurons from a cubic millimetre of mouse visual cortex, with in-vivo responses to a shared movie, a fitted digital-twin model [10], orientation labels for 5,287 of them and receptive-field labels for 11,326 (Fig. 1). The decoder trained on half the neurons reads the other half. Across brains we use the Allen Brain Observatory [11]: 33 mice recorded under identical stimuli. The decoder trained on some animals reads neurons of animals it never saw, which succeeds only when a sampled population mixes animals, and only with grating relations.

Averaging the correlations by orientation class shows that the relations between classes are almost circulant, which fixes the classes only up to a rotation or reflection of the orientation circle. A consistent departure from that symmetry fixes the absolute frame and makes the readout possible.

2 Methods

2.1 Data

MICrONS. The node table released with Ding et al. [8] holds 12,894 neurons co-registered to the electron-microscopy volume, across V1, RL, AL and LM: the 13,952 matched excitatory neurons after the authors’ restrictions (one of the 14 two-photon scans dropped for compromised optics, a threshold on the co-registration match, and one functional unit per neuron, the one with the higher oracle score). The paper does not report the count after these steps; 12,894 is the row count of the table. Each neuron has a trial-averaged in-vivo response to the oracle natural-movie clips (120 bins), shown in every scan; a digital-twin response to a shared movie (4,999 bins, which we compress to 512 principal components; the Gram is preserved to $r > 0.999$); an in-vivo preferred orientation with a global orientation selectivity index (gOSI); and a receptive-field centre from the twin (spike-triggered-average fit, screen coordinates in $[-1, 1]$). We use orientation, as a continuous angle, for the 5,287 neurons with $\text{gOSI} \geq 0.25$, and receptive-field position for the 11,326 neurons whose twin test correlation is ≥ 0.2 . The orientation labels are in-vivo measurements on both substrates; the receptive-field labels come from the twin’s fit on both substrates, so every receptive-field result, in vivo included, scores relations against model-derived

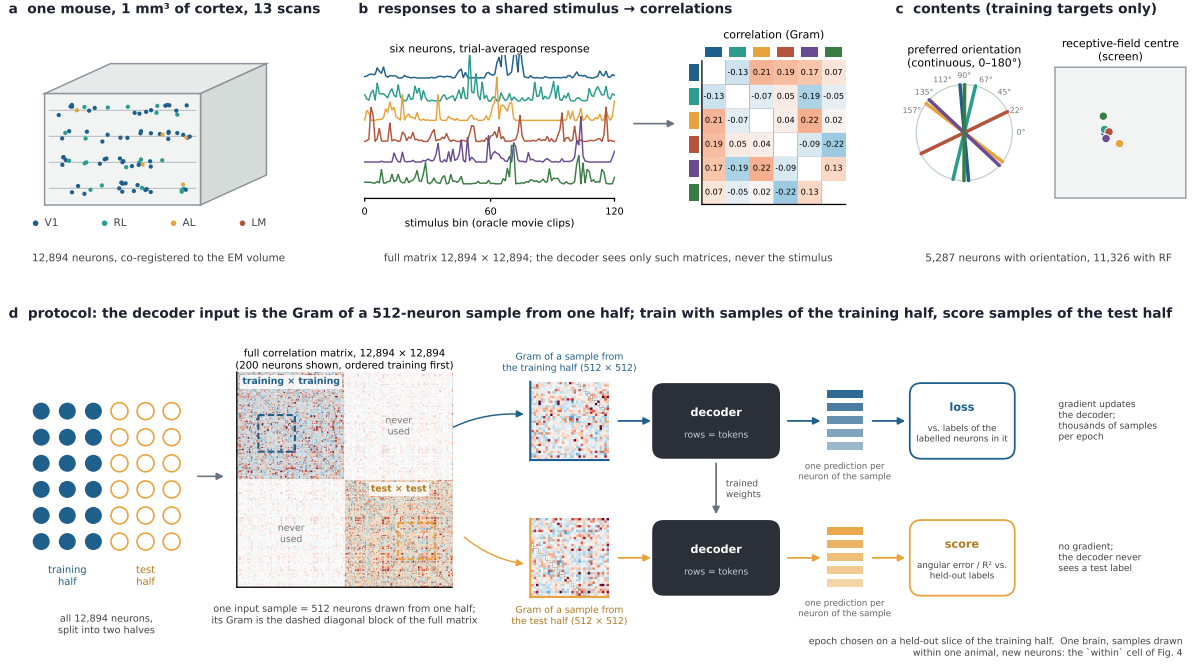


Figure 1: How the MICrONS task is built. (a) One mouse, 13 of the 14 two-photon scans of a cubic millimetre of visual cortex (areas V1, RL, AL, LM), 12,894 neurons co-registered to the electron-microscopy volume. (b) The relational substrate. Each neuron is its trial-averaged response vector to a stimulus all neurons saw (Methods); shown are six real neurons and their correlation matrix. The oracle clips were shown in all 13 scans, so the matrix is defined across scans. The decoder sees only such correlations, never the stimulus. (c) The contents of the same six neurons: preferred orientation (in vivo, a continuous angle, 5,287 neurons with $\text{gOSI} \geq 0.25$) and receptive-field centre (digital-twin fit, 11,326 neurons). These are training targets only. (d) The protocol. Neurons are split into a training half and a test half; the full correlation matrix, ordered training-first (a real 200-neuron excerpt), is used only in its training×training and test×test blocks. A sample is 512 neurons drawn from one half; its 512 × 512 Gram (dashed) is the decoder’s entire input, one row per neuron, one prediction per row. Labels enter only the loss, never the input, and the halves share no neuron (Methods).

positions (Section 4). Same-scan pairs correlate about 50% more strongly than cross-scan pairs at matched receptive-field distance. The content of the correlations is the same across scans, and all populations mix scans. Only 2% of the receptive-field label variance lies between scans and 9% between areas, so a decoder that read scan or area identity from the Gram could reach at most that much R^2 .

Allen. From Visual Coding 2P [11] we take the 36 VISp excitatory containers with at least 150 cells significantly responsive to drifting gratings: 33 mice, 9,281 cells, of which 5,529 carry the static-grating orientation label used here; unlabelled cells stay in the populations and receive no loss, so a held-out-animal split has 6,706 training cells and 2,575 test cells in total. Relations come from session A, either responses to natural movies shown to every mouse (4,500 bins) or the 40 blank-subtracted condition means of drifting gratings (8 directions $\times$ 5 temporal frequencies). Orientation labels come from static gratings in session B (six orientations 30° apart, $\text{gOSI} \geq 0.25$, $p < 0.05$), so labels and relations use different stimuli. Receptive-field centres come from locally sparse noise in session C (1,504 cells with a significant fit, 23 to 112 per mouse).

2.2 Relations

Each neuron’s response vector is centred and normalised; the relation between two neurons is the cosine of their vectors, a signal correlation. For a population of n neurons the Gram is the $n \times n$ matrix of these values, standardised by the global off-diagonal mean and standard deviation, with the diagonal zeroed. We call this standardised correlation matrix the Gram throughout.

2.3 Decoder

Each row of a population’s Gram, together with three row statistics (the mean, standard deviation and third moment of the row), is projected to a token; L pre-norm transformer blocks [12, 13] with H -head attention update the tokens, the Gram entering the attention logits as an additive bias; a linear head reads out, per token, the unit vector $(\cos 2\theta, \sin 2\theta)$ of the preferred orientation θ or an (x, y) receptive-field position. The loss is the mean squared error on the labelled tokens only; unlabelled neurons stay in the population and receive no loss. The standard configuration is width 512, 8 layers and 17M parameters with populations of 512 neurons on MICrONS, and width 256, 4 layers and 2.2M parameters with populations of 256 neurons on Allen. Other sizes are in Appendix A. Training draws thousands of random populations per epoch.

2.4 Protocol

Split. MICrONS neurons are halved at random. Training populations are sampled inside the training half and test populations inside the test half, so no test neuron appears in a training population (Fig. 1d). A slice of the training half is held out to choose the epoch. The test half never influences selection.

Regimes (Allen). Two independent choices define a run. The split decides where test neurons come from: the training animals, each halved into training and test neurons, or held-out animals. The population decides what a sample is: neurons drawn from one animal, so that the Gram is a within-circuit matrix, or from several animals, so that most entries are between-animal correlations. We call the four cells **within**, **pooledwithin**, **cross** and **pooledcross**. Held-out regimes select the epoch on further held-out mice. Every held-out cell is run on three mouse splits and every training-animal cell on three neuron splits.

Metrics. Orientation is scored by the mean absolute angular error between the decoded angle, $\frac{1}{2} \text{atan2}$ of the output, and the label, taken modulo 180° , so it lies between 0° and 90° . A decoder whose output is independent of the label produces errors spread evenly over that range when either the labels or the outputs are uniform on the orientation circle, so chance is 45° . With the MICrONS label distribution the best constant prediction scores 41.3° (Appendix B). Both lines are drawn in the figures. Allen labels sit on a 30° grid, so a decoder that recovered every true preference exactly would still disagree with them by 7.5° on average. That is the floor for Allen errors. A frame check reports the error after the best single rotation or reflection of all decoded angles. If it is lower than the raw error, the decoder recovered the structure but not the frame. Receptive fields are scored by R^2 against a baseline of 0.

2.5 Symmetry analysis

To expose the symmetry of the relations we average the Gram into a $K \times K$ class-Gram, the mean relation between neurons of two orientation classes, project it onto its circulant part, and ask which relabellings a held-out half's class-Gram is compatible with (all $K!$ relabellings, Frobenius distance). A relabelling that differs from the truth by a rotation or a reflection of the orientation circle is an element of the dihedral group D_K . The analysis reports whether the truth is singled out absolutely or only modulo D_K .

3 Results

3.1 Per-neuron content from the correlation matrix alone

On MICrONS the decoder reads orientation and receptive-field position from the Gram of 512 neurons of the test half (Fig. 2, Table 1). The twin substrate is cleaner, since the twin removes trial noise, and receptive fields carry more than orientation. The predictions are in the true frame: the frame check lowers the error by less than 0.1° in every run. About half the neurons are decoded to within 15° (0.46 in vivo, 0.54 on the twin), but the error is not spread evenly over the orientation circle: the decoder resolves the cardinal axes and little else (Appendix B, Fig. 3). Asked only which cardinal axis a neuron is closer to, it is right for 87% of all neurons on the twin and 79% in vivo (majority answer 56%; averaged over the eight true-orientation bins, 83% and 74%), and for the neurons whose preference lies within 15° of an axis, for 94% and 86% (neuron splits 1 and 2). Those near-axis neurons are 56% of the population. Sorted into four classes at 0° , 45° , 90° and 135° , 66% (twin) and 59% (in vivo) land in the right one against a majority class of 40%. The misses are almost all obliques. Within 15° of a cardinal axis the error is 12° on the twin and 19° in vivo; within 15° of an oblique, 36° to 39° over both substrates, near chance (Fig. 3). Both readouts are properties of individual neurons, not of the scan: centring predictions and labels per scan leaves the receptive-field R^2 unchanged and the orientation error within 0.2° (Appendix B). Test Grams from stimulus bins the decoder never trained on add about 0.5° of error on the twin and 8.5° in vivo, where a Gram over 60 movie bins is partly specific to those bins (Appendix B).

3.2 Why the absolute frame is recoverable

The class-Gram of orientation is close to circulant (Fig. 4): neighbouring orientations correlate, orthogonal ones anti-correlate, and the pattern repeats around the circle. A circulant matrix is unchanged by every rotation and reflection of the orientation circle, so that part of the relations fixes the classes only up to the 16 elements of D_8 and cannot say which class is 0° . Whatever fixes the frame must be a departure from circulant structure, at the class level or finer. At the class level that residual holds 29% of the class-Gram variance in vivo and 19% on the twin over the whole volume (30% and 16% in V1), and it is signal: it correlates at 0.94 between random

Table 1: MICrONS, per-neuron decoding on the test half. Relations only: the 17M decoder on the Gram of 512-neuron populations, mean $\pm$ s.d. over 3 seeds. Balanced: the error averaged over eight equally weighted true-orientation bins (neuron split 2). Stimulus-known linear readout: a linear map from each neuron’s own movie response vector to its content, a reference that uses the stimulus alignment the relational decoder discards, not a label-noise ceiling (Appendix B). Chance: an uninformed decoder (45°) and the best constant orientation given the label distribution (41.3°).

content	substrate	relations only	balanced over orientations	stimulus-known linear readout	chance / best constant
orientation (angular error)	in vivo	$25.4^\circ \pm 0.5^\circ$	27.9°	20.2°	$45^\circ / 41.3^\circ$
orientation (angular error)	twin	$20.2^\circ \pm 0.4^\circ$	24.5°	15.8°	$45^\circ / 41.3^\circ$
receptive field (R^2)	in vivo	0.234 ± 0.007	—	0.34	0
receptive field (R^2)	twin	0.480 ± 0.010	—	0.68	0

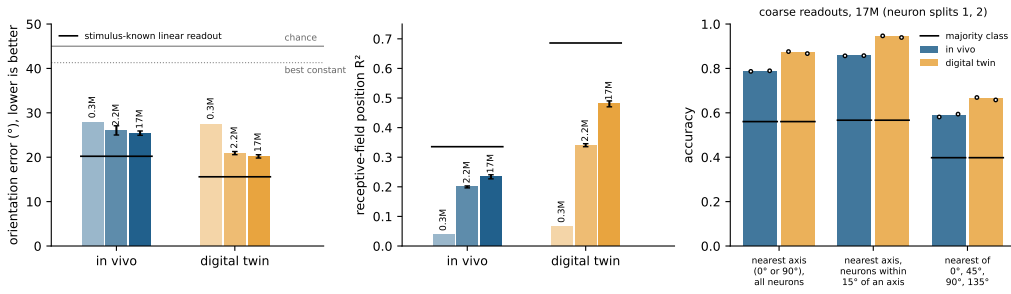


Figure 2: MICrONS: per-neuron content decoded from the population correlation matrix alone. Left: mean angular error of the decoded preferred orientation on held-out neurons of the same animal, for decoders of 0.3M, 2.2M and 17M parameters; error bars are the s.d. over 3 seeds; grey line: the 45° error of an uninformed decoder; dotted: the best constant, 41.3° ; black marks: the stimulus-known linear readout of each neuron’s own responses. Middle: R^2 of the decoded receptive-field centre, same marks. Right: coarse readouts of the same decoder on neuron splits 1 and 2: whether the decoded angle falls on the correct cardinal axis, for all neurons and for the neurons whose preference lies within 15° of an axis, and whether it falls in the correct one of four classes at 0° , 45° , 90° and 135° ; black ticks: majority class. The decoder sees only the standardised correlation matrix of 512 sampled neurons.

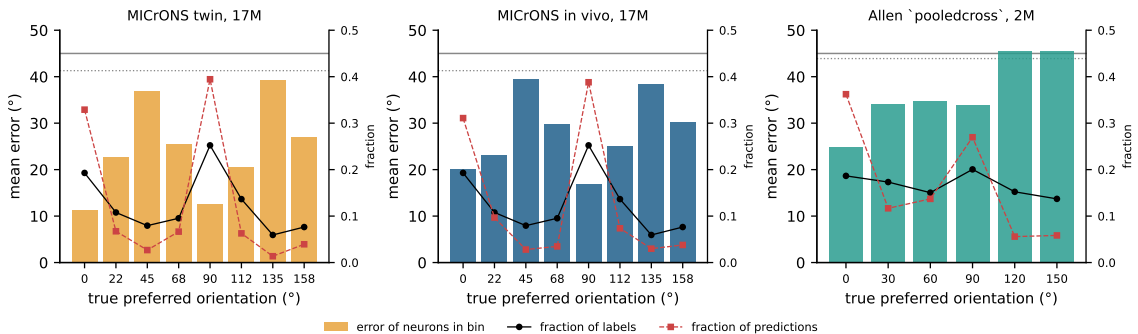


Figure 3: The readout is cardinal. Mean angular error of the neurons in each true-orientation bin (bars; grey line: 45° chance; dotted: best constant), with the fraction of labels (black) and of predictions (red) per bin, for the twin and in-vivo MICrONS decoders (neuron split 2) and the Allen `pooledcross` decoder. Cardinal orientations are decoded well, obliques near chance, and most predictions fall on 0° or 90° .

halves of the neurons on both substrates, at 0.84 (in vivo) and 0.99 (twin) between disjoint halves of the stimulus bins, and at 0.99 with the residual computed on class-balanced subsamples, so it is neither a finite-stimulus artefact nor a consequence of the uneven number of neurons per orientation. Its form is one-peaked, not bi-cardinal: in vivo, neurons preferring orientations within about 45° of 90° correlate more with each other than the circulant baseline predicts and the rest less, and on both substrates the same-class excess at 90° exceeds that at 0° (V1: $+0.036$ against -0.011 in vivo, $+0.014$ against $+0.006$ on the twin, where the asymmetry is the smaller), and its shape differs between areas (residual correlation V1-RL 0.77, V1-AL 0.31).

Four observations narrow down what fixes the frame. A fifth points to a specific anisotropy, which a model population then tests.

Priors do not explain the frame. The decoder recovers the absolute frame (Section 3.1), while predictors that use only the distribution of labels, the best constant orientation or each scan’s or area’s mean, reach 39° to 41° .

The structure is recoverable with no frame in the labels. A decoder trained with each population’s targets rotated at random, and the loss taken after the best alignment of predictions to targets, places neurons to 19.9° on the twin and 25.5° in vivo after one global alignment, the same as the standard decoder with the frame (20.2° , 25.4°). The per-neuron structure needs no frame anywhere in the supervision. This does not show that the circulant part alone carries the structure, since the residual is still in that decoder’s input. Its raw error, 25° and 31° against 45° for an arbitrary frame, shows that it fixes a consistent frame from the data even when the loss does not ask for one.

The frame is not the label density and not the stimulus sample. With every training and test population drawn with equal numbers of neurons from each of the eight orientation classes, the decoder reaches 22.0° on the twin and 26.6° in vivo (two seeds each), with no gap between raw and aligned error. It is better than the standard decoder scored with the same equal weighting (24.5° , 27.9°). The frame also survives training and test relations built from disjoint halves of the stimulus, and it survives removing the three row statistics from the input (20.3°), while the statistics alone fix only a weak frame (37.7° ; Appendix B).

The errors have the shape of the symmetry breaking. They are 11° to 12° near the cardinal axes and 37° to 39° in the two oblique classes on the twin, and reweighting the loss so that obliques count as much as cardinals leaves them unchanged (Appendix B). The oblique ordering is not in the relations at single-neuron resolution.

In V1 the neurons of the dominant orientation correlate most strongly with each other. Define the anisotropy factor as the mean correlation between two neurons of the same orientation class, at the class where it peaks, divided by its mean over the other seven classes. In V1, where 28% of neurons fall in the 22.5° class centred on 90° , the factor peaks at 90° on both substrates (1.28 on the twin, 1.77 in vivo; Table 2), and the values are unchanged on class-balanced subsamples, so the signature is in the correlations, not in the counts. The whole volume, which the decoder reads, peaks at 90° as well. In RL the peak is at 112.5° on both substrates against a label mode of 0° ; in AL it coincides with the label mode on the twin but sits at 135° in vivo, a disagreement between the substrates in the smallest area. The coincidence therefore holds in V1, which supplies 73% of the labelled neurons, so the volume the decoder reads inherits V1’s peak; it does not hold in the two smaller areas. The quantity is a signal correlation over trial-averaged responses, and on the twin over noiseless ones, so the anisotropy is one of tuning similarity, not of trial-to-trial co-fluctuation. Cardinal over-representation and its variation between areas are established features of mouse visual cortex [14, 15], and the preferential connectivity of like-tuned neurons [16] is one route by which their responses could become more similar.

A model population tests the candidate. We could not isolate the anchor by subtraction, because removing the class-level residual from every Gram with the true classes injects the classes into the Gram (Appendix B), so the frame is located by exclusion and by a model (Appendix B, Table 6). Model neurons with von Mises tuning, a movie-like stimulus that covers orientation

Table 2: Label mode and anisotropy peak per area (MICrONS, 22.5° classes). Label mode: the class with the most neurons and its fraction. Peak: the class with the largest mean same-class correlation and the anisotropy factor (peak over the mean of the other seven classes). LM has 10 labelled neurons and is not given its own row; the volume row includes it.

area	neurons	label mode	peak, twin	peak, in vivo
V1	3,858	90° (0.28)	90° (1.28)	90° (1.77)
RL	1,125	0° (0.24)	112.5° (1.22)	112.5° (1.68)
AL	294	0° (0.32)	0° (1.49)	135° (1.69)
whole volume	5,287	90° (0.25)	90° (1.26)	90° (1.81)

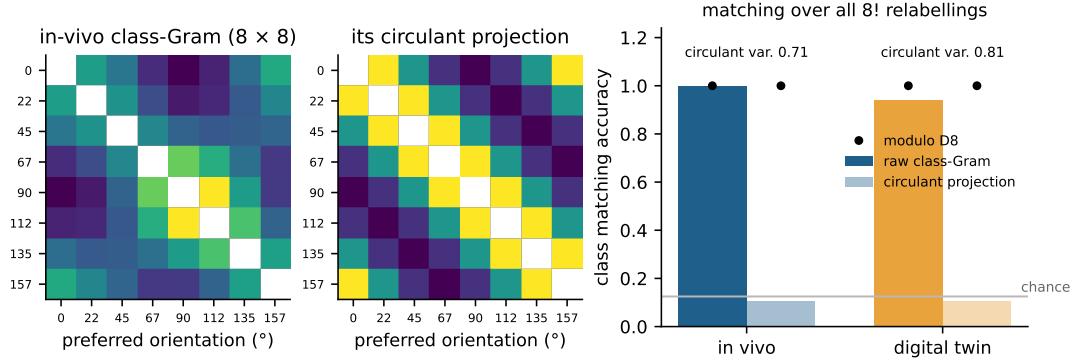


Figure 4: What fixes the absolute orientation frame. Left: mean correlation between orientation classes in vivo (diagonal masked). Middle: its circulant projection. Right: matching a held-out half’s class-Gram to the reference over all 8! relabellings. With the raw matrix the true labelling wins. After the circulant projection, absolute accuracy collapses to chance while accuracy modulo D_8 (dots) stays at 1.0, since a circulant matrix is invariant under D_8 . The bars sit at or slightly below the chance line because the matching then picks one of the 16 equivalent labellings at random. The panel shows what the residual has to supply.

evenly, and the same pipeline give a class-Gram circulant to within noise, and the decoder trained on it recovers the ordering, 30° after one global alignment, but no frame, 54° raw. Sharper cardinal tuning and a cardinal excess of neurons fix the frame in at most one seed of five, alone, together, or at any strength tested; an excess of cardinal orientations among the stimulus frames does in three of five. Adding only the anisotropy the data show, a shared stimulus-driven response component among like-tuned neurons peaked at one orientation, with uniform counts and an even stimulus, fixes the frame in every seed at every strength run (15 of 15, 2° to 4° raw), from a residual weaker than the twin V1’s (0.05 of class-Gram variance against 0.16) to one between the twin’s and the in-vivo V1 value (0.23 against 0.30); no strength tested fails, so no threshold was located; the weakest setting is below cortex on the residual fraction and above it on the anisotropy factor (1.67 against 1.28 on the twin), so the model is not calibrated to cortex in a single ordering. The model residual has the one-peaked form of the cortical one and correlates 0.39 to 0.48 with it in vivo, against 0.94 for the cortical residual with itself across neuron halves. The model is far easier to decode than cortex (2° against 20° at a similar residual), so it shows that the anisotropy suffices, not that it accounts for the cortical error. The cortical anisotropy is not inherited from the stimulus, since it reproduces across disjoint halves of the stimulus and differs between areas that saw the same movie.

3.3 Orientation transfers between animals through mixed populations

With natural-movie relations, orientation is nearly absent from Allen correlations: same-orientation pairs correlate 0.005 more than orthogonal pairs, against 0.09 in MICrONS, and every test is at chance. The Allen results below therefore use grating relations, a designed set

Table 3: Allen, mean angular error of the decoded orientation in the four regimes (2.2M decoder, 256-neuron populations; 128 for single-animal populations), three splits per cell (of neurons within each animal for the top row, of animals for the second row). Last row: the same decoder on held-out animals with each neuron’s 40-condition response vector as its token instead of its Gram row, the stimulus-known reference. Chance 45° , best constant given the Allen label distribution 43.9° ; label grid 7.5° . Means over the three splits in parentheses.

test neurons from	single-animal populations	mixed populations
the training animals	$42.5^\circ / 42.9^\circ / 42.1^\circ$ (42.5°)	$35.3^\circ / 37.2^\circ / 36.1^\circ$ (36.2°)
held-out animals	$43.1^\circ / 43.1^\circ / 42.7^\circ$ (43.0°)	$35.2^\circ / 37.0^\circ / 38.4^\circ$ (36.9°)
held-out animals, stimulus-known reference	$32.8^\circ / 33.6^\circ / 34.9^\circ$ (33.8°)	$32.4^\circ / 33.7^\circ / 34.3^\circ$ (33.5°)

of 40 conditions over an orientation grid, and do not replicate on naturalistic relations. With gratings the signal is present (0.05 within a mouse, 0.044 across mice) and the four regimes separate (Fig. 5, Table 3). Single-animal populations carry a weak but consistent signal: all six runs lie 1.9° to 2.9° below chance and 0.3° to 1.8° below the best constant of their own test set (43.3° to 43.9° per split), whether or not the animal was seen in training, and on every split the mixed populations of the same regime are 4.3° to 7.9° better (mixed populations of 128 neurons, matching the single-animal size, score 36.0° ; Appendix B). Mixed populations are 6.6° to 9.8° below chance in both rows, at the same level, so no loss from transfer to a new brain is detectable at three splits. As on MICrONS the readout is cardinal: over the three splits the decoder names the right cardinal axis for 62% of all cells and 68% of the cells labelled 0° or 90° (majority 51% and 52%; 61% averaged over the six label bins), cells labelled 0° are decoded to 25° and those labelled 90° to 34° , cells labelled 120° or 150° are at chance, and the four-class readout is at the majority class (Fig. 3). What matters is whether the Gram contains between-animal correlations. These exist only because all mice saw the same 40 conditions, and they measure how similar two tuning profiles are. The decoder still never sees which condition is which, so the absolute frame must again come from the population’s relational structure. Orientation is thus readable from a correlation geometry that spans animals, and only weakly from relations inside one circuit. Two controls say why. The limit is not the neurons: with each neuron’s own 40-condition response vector as its token, the same decoder on the same single-animal populations reaches 33° to 35° , as with mixed populations (Table 3, last row). The limit is the training pool: mixed populations drawn from only 280 or 560 training cells, one or two animals’ worth, score 40.5° and 40.7° , near chance, against 36.0° from the full pool of 6,706 at the same population size of 128 (Appendix B). An animal has about 280 cells, about 170 of them labelled, and half of them are training cells in the within-animal split, so a decoder restricted to within-animal correlations is trained on about 140 cells per animal, too few to learn from whatever those correlations contain. The same limit appears on MICrONS: a decoder trained on 140 neurons of one scan and tested on the other scans scores 38° to 44° on the twin and 37° to 39° in vivo (three scans; Appendix B). Where one animal supplies thousands of training neurons, single-scan populations are decoded to 32.0° on the twin and 31.1° in vivo (128 neurons; Appendix B), against 22.3° and 25.8° for mixed-scan populations of the same size (three seeds each). Relations inside one circuit do carry orientation, more weakly than relations that span circuits.

3.4 Receptive fields: what crosses animals is the animal’s screen position

Correlation falls with receptive-field distance within a mouse (0.127 at $< 5^\circ$ to 0.068 beyond 40°) and across mice (0.049 to 0.011). Decoding absolute screen coordinates on held-out mice gives R^2 that varies with the split, from 0.29 to -0.45 for single-animal populations and from 0.22 to -0.25 for mixed ones. Decomposing the saved predictions explains the variation (Fig. 6, Appendix C). The correlation between predicted and true mean position of each held-out mouse, pooled over 27 mouse-level predictions, is 0.36 to 0.71 (permutation $p \leq 0.03$). The correlation

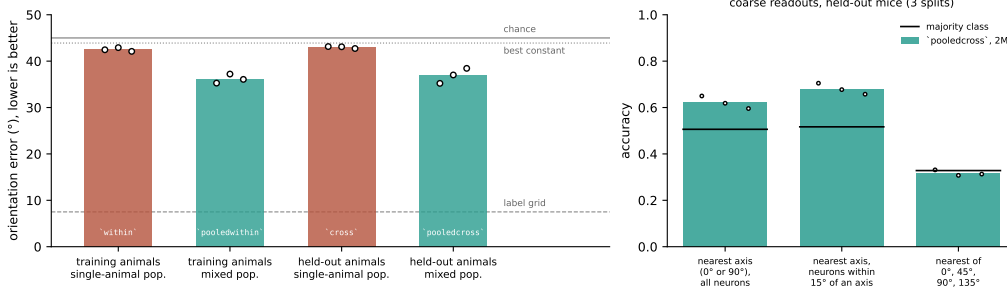


Figure 5: Allen, 33 mice, grating relations: orientation is readable only from populations that mix animals. Left: mean angular error; bars: mean over splits; points: the three splits; solid line: 45° chance; dotted: the best constant, 43.9°; dashed: the 7.5° label grid. Single-animal populations are 1.9° to 2.9° below chance whether the animal was seen in training or not; mixed populations are 6.6° to 9.8° below it in both rows. Right: coarse readouts of the mixed-population decoder on the held-out mice (three splits): correct cardinal axis for all cells and for the cells labelled 0° or 90°; the third bar, nearest of four classes 45° apart, maps each 30° label to one class (30° and 60° to 45°, 120° and 150° to 135°) and is shown for comparison with MICrONS; black ticks: majority class.

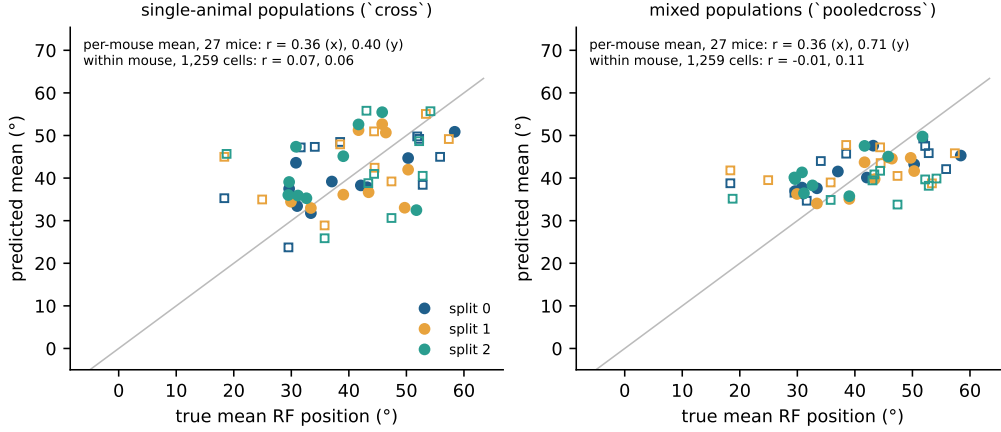


Figure 6: Allen receptive fields: what crosses animals is each animal's screen position. Predicted versus true mean receptive-field position of each held-out mouse (27 points over 3 splits; x squares, y dots), from decoders trained on single-animal (left) or mixed (right) populations. The per-mouse mean is recovered; the position of a neuron relative to its mouse-mates is not.

between predicted and true position of a neuron relative to its mouse-mates is -0.01 to 0.11 over 1,259 cells. The decoder recovers where an animal's imaged patch looks, not which neuron within it looks where. R^2 on absolute coordinates then depends mostly on whether the between-mouse spread of a particular test set is reproduced at the right scale.

The within-mouse layout is present in the correlations, since same-mouse correlation falls with distance. It is not learned because each mouse contributes only 23 to 112 labelled cells. With 11,326 labels in one animal, MICrONS lets the same decoder reach neuron-level receptive fields at $R^2 = 0.48$.

4 Discussion

The correlation matrix of a few hundred cortical neurons, with the stimulus unknown and no neuron labelled, determines each neuron’s preferred orientation to a mean error of 25° in vivo and 20° on the digital twin, against 45° for an uninformed decoder and 41° for the best constant guess. The readout is cardinal: near-cardinal neurons to 12° to 19° , obliques near chance. It determines each neuron’s receptive-field position to $R^2 = 0.23$ in vivo and 0.48 on the digital twin. The symmetry analysis constrains the mechanism: the relations fix the class structure up to the symmetries of the content, and a departure from that symmetry fixes the frame. That departure is a property of the correlations themselves: it is not the uneven number of neurons per orientation and not the particular stimulus sample. In V1 the neurons preferring the over-represented orientation have the most similar responses to one another, whether or not the sampled population is imbalanced. A direct ablation of that feature in the data is not available, but a model population with that anisotropy and nothing else recovers the absolute frame at every strength tested, while models with sharper or more numerous cardinal neurons do not.

Across animals the two contents behave differently. Orientation is carried by the relations between tuning profiles, which exist between any two neurons that saw the same conditions, in the same brain or not. From the correlations inside a single scan it is readable only weakly, at 31° to 32° on MICrONS, and inside a single Allen animal barely, where 140 training cells are too few to learn from. Receptive-field position is carried at the level of an animal’s patch, and the neuron-level layout needs more labels per animal than these recordings provide. Both statements concern what correlations under a shared stimulus contain, and both come from a protocol in which the test neurons, and in the held-out regimes the test animals, never touch training or model selection.

Limits. MICrONS is one animal, and its training and test halves share recording sessions. The Allen held-out-animal cells are the cleaner generalisation test, but their variance across mouse splits is large, and they exist only with grating relations, a designed 40-condition set on which two tuning profiles can be compared, not with the naturalistic relations of the MICrONS result. The readout is cardinal: the obliques are not resolved on either dataset, so the mean error understates the precision at the cardinals and overstates it elsewhere. No experiment removes the residual from the decoder’s input, so the assignment of the ordering to the circulant part and the frame to the residual is inferred from the symmetry argument and the model, not measured. The mechanism is established by exclusion and by a model, not by removing the candidate anisotropy from cortical data, since the direct ablation we tried injected the labels it was meant to remove (Appendix B). Orientation labels are noisy in both datasets (in vivo and twin labels agree on 70% of MICrONS neurons at 8 classes; static and drifting gratings on 45 to 67% of Allen cells), and the Allen labels are quantised to 30° , which caps every labelled and label-free number alike. The twin substrate is a model fitted to the same recordings, so the in-vivo results carry the claim about cortex and the twin results show what cleaner correlations would give; in AL the two substrates place the anisotropy peak 135° apart, though the in-vivo peak is within noise of three other classes at 294 neurons. The receptive-field labels come from the twin for both substrates, so the twin receptive-field result compares a model’s correlations with the same model’s receptive fields, and the in-vivo result scores cortical correlations against model-estimated positions. The orientation labels are in-vivo measurements throughout.

The symmetry result is an instance of a general expectation: a representation that is equivariant to a group has a group-fixed part, identical across individuals up to relabelling, and an individual remainder [17]. The matching-over-relabellings test is the discrete form of unsupervised alignment of similarity structures across individuals by optimal transport [18]. This paper adds the measurement in cortex, at the level of single neurons, and the finding that the absolute readout is possible and rests on a remainder that belongs to the correlations, not to the sample of neurons or stimuli.

Code and data. TODO: repository URL. All data are public releases.

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A Scaling and population size

More capacity helps where labels are plentiful and not where they are scarce (Fig. A1). Twin receptive fields go from 0.34 to 0.48 to 0.51 R^2 between 2.2M, 17M and 57M parameters. Orientation error is flat from 2.2M upward on both MICrONS substrates (20.9°, 20.2°, 19.9° on the twin at 2.2M, 17M and 57M) and on Allen (35.2°, 36.5°, 36.7°); only the 0.3M model is worse (27.5° in vivo and 27.8° on the twin). Every model above 2M memorises its training neurons within 10 to 20 epochs and is early-stopped on the held-out selection slice. Population size beyond 256 to 512 neurons changes orientation error by at most 2° in either direction, while twin receptive fields keep gaining slowly (Fig. A2). Recomputing each training population’s Gram from a random subset of the stimulus conditions, a label-free augmentation, gives no consistent gain on the Allen error (35.2°, 33.5°, 37.7° and 38.0° with 100%, 75%, 50% and 30% of the conditions at 2.2M; 36.5°, 34.3° and 35.6° with 100%, 85% and 50% at 17M; single runs, against a split-to-split spread of 3°) and is not used; zeroing random Gram entries during training was dropped for the same reason. Split-to-split variance on held-out mice exceeds seed variance (Fig. A3).

B Controls

Controls on the MICrONS standard configuration (17M, 512-neuron populations, digital twin unless stated; Table 5).

What the decoder reads. With the three row statistics as the only input per neuron and no relational term, the error is 37.7°, raw and after alignment alike, so the scalars fix a weak frame on their own; with the Gram row alone and no statistics it is 20.3° on the twin and 26.5° in vivo, raw and aligned, so the relations carry the frame and the scalars are dispensable. (They are not a selectivity confound: their correlations with distance to the nearest cardinal axis are below 0.12 in magnitude and with gOSI below 0.16.) With the statistics plus the Gram entering only as an attention bias, which makes the decoder equivariant to the order of the neurons, the error is 19.9° over three seeds, the same as the standard decoder (20.2°), whose input layer also projects the raw Gram row in sample order.

Reference decoders that see the activity. To measure what the relations discard, the same transformer was trained with each neuron’s token built from its response vector instead of its Gram row, so that it sees the stimulus-aligned activity the Gram is computed from (Table 4). It does better than the relational decoder on both contents and both substrates. Adding the Gram to it as an attention bias changes nothing. A linear readout of each neuron’s own response vector, with no population context, does as well as the activity transformer, so the activity decoder is a per-neuron tuning readout whose only advantage over the relational decoder is the alignment of the responses to the stimulus. On orientation the relations retain about 80% of the gain over the best constant that the aligned activity gives, and on receptive fields about 60%; the twin receptive-field gap is inflated by the twin’s labels and responses coming from the same model. In exchange the relational decoder does not depend on the stimulus alignment. With test responses from movie frames the decoder never trained on, the activity decoder has no valid input, since its coordinates now index different frames, and it scores at chance or worse, while the relational decoder scores 33.8° in vivo and 20.7° on the twin. This shows which inputs each readout can be applied to, not which readout is better. The activity decoder transfers across animals under a shared stimulus protocol (Section 3.3) but not across stimuli. The relations do both, because a correlation does not depend on which frames it was computed from, only on both neurons having seen the same ones. A decoder given the response vectors with the stimulus bins shuffled, the same shuffle for every neuron of a population, is blind to the stimulus but free to compute any

relation among the neurons. It is no better than the Gram decoder on orientation and far below it on receptive fields (Table 4), but it has to discover the relations under a fresh permutation for every population, so the comparison does not show whether richer statistics than the pairwise Gram exist.

Table 4: Reference decoders that see the activity (each neuron’s stimulus-aligned response vector), 17M unless linear, 512-neuron populations. Orientation as mean angular error (chance 45°), receptive field as R^2 ; mean $\pm$ s.d. over three seeds where given, otherwise seed 0.

decoder input	ori. in vivo	ori. twin	RF in vivo	RF twin
Gram row (this paper)	$25.4^\circ \pm 0.5^\circ$	$20.2^\circ \pm 0.4^\circ$	0.23 ± 0.01	0.48 ± 0.01
activity, transformer	$21.5^\circ \pm 0.4^\circ$	$15.0^\circ \pm 0.1^\circ$	0.37	0.80
activity + Gram as attention bias	22.2°	15.0°	–	–
activity, linear, no population context	$20.2^\circ \pm 0.2^\circ$	$15.8^\circ \pm 0.1^\circ$	0.34 ± 0.01	0.68 ± 0.01
activity, transformer, disjoint stimulus bins	49.5°	41.0°	–	–
activity, bins shuffled per population, transformer	$26.3^\circ \pm 0.6^\circ$	$27.8^\circ \pm 6.5^\circ$	0.04	0.08
activity, bins shuffled, linear, no population context	40.3°	40.2°	–	–

Cross-stimulus test. Training and selection Grams built from one random half of the stimulus bins and test Grams from the other half give 20.7° on the twin (2,500 bins per half, raw responses rather than principal components; three seeds, 20.3 to 21.2) and 33.8° in vivo (60 bins per half; 33.4 to 34.2), against 20.2° and 25.4° with shared bins. A control with the same 60-bin half on both sides gives 25.3° , so the in-vivo increase comes from the stimulus sample, not the smaller Gram: a correlation matrix over 60 movie bins partly reflects which frames it was built from, and one over 2,500 bins does not. The relational structure the decoder reads is stable across stimuli once the stimulus sample is large.

Scan and area. The MICrONS neurons come from 13 scans in four areas, same-scan pairs correlate more strongly than cross-scan pairs, and the scans look at different parts of the screen, so a decoder that read scan identity from the Gram could predict each scan’s mean receptive-field position and score a positive R^2 without knowing any neuron’s position. The Allen decoder does exactly that (Section 3.4), so we checked MICrONS the same way, centring predictions and labels per scan and per area before scoring. The receptive-field result holds: on a fourth seed of the Table 1 configuration, run with saved predictions, R^2 is 0.499 absolute and 0.498 within scan on the twin, 0.249 and 0.253 in vivo, and 0.47 (twin) and 0.22 (in vivo) within area; the fourth seed’s twin value is two standard deviations above the three-seed mean of Table 1. Only 2% of the label variance lies between scans and 9% between areas, so a scan-identity readout could reach at most 0.02. For orientation, a predictor that assigns every neuron its scan’s or area’s mean orientation scores 39° , and removing a separate rotation per scan or area changes the decoder’s error by less than 0.2° . Both MICrONS results are neuron-level readouts within a scan. Decoders trained on V1 alone transfer to the other areas unevenly across seeds (Table 5): to AL the aligned error is 32° in all three seeds but only one seed lands on a rotated frame, and to RL one seed transfers and two do not. The single-seed rotation of 78° reported in an earlier draft did not replicate, and no claim rests on these rows; the RL row uses 512-neuron populations and the V1 and AL rows 128. With class-balanced training populations, V1 itself is decoded 9° worse than with the standard training set, against 1.8° for class balancing of the whole volume, because V1’s cardinal-heavy test set rewards the prior that balancing removes.

Class balance and priors. Preferred orientations are uneven in MICrONS (19% of labelled neurons within 11° of 0° , 25% of 90° , 6 to 10% in each oblique bin), so a decoder could gain from the distribution alone. It gains little: the best constant prediction has an error of 41.3° , a predictor that assigns every neuron the circular mean of its scan 39.0° and of its area 38.7° ,

against 20.3° for the decoder on the twin and 24.6° in vivo (neuron split 2). Training and testing on populations drawn with equal numbers of neurons from each of the eight classes (64 per class) gives 22.0° on the twin and 26.6° in vivo (two seeds: 22.4, 21.5; 26.5, 26.7), raw and after global alignment alike, so the absolute frame does not depend on the class imbalance. The per-area errors are within 3° of each other: the frame is read from a volume-wide sample, so an area whose label mode differs from the volume’s peak is not mis-framed. On Allen the labels are nearly flat (14 to 20% per orientation), the best constant is 43.9° , and the balanced error is 36.4° against 35.7° raw on the saved per-neuron predictions of held-out-animal split 0, which average 32 population draws per neuron (Table 3 reports the population-level error of the same run, 35.2°). The per-bin errors show, on both datasets, that the readout is cardinal: neurons preferring 0° or 90° are decoded to 11° to 12° on the twin and to 25° and 34° on Allen, oblique ones to 37° to 39° on the twin and at chance on Allen, and 63 to 72% of all predictions fall on the two cardinal axes (Fig. 3). The relations say how close a neuron is to a cardinal axis, the anisotropy of Section 3.2 seen from the decoder’s side, and much less about which oblique it prefers.

Table 5: Controls, mean angular error on held-out neurons (chance 45°), 17M decoder on 512-neuron populations unless stated; single runs unless a seed count is given. Where two numbers are given, the second is the error after the best global rotation or reflection (rows marked *aligned*) or averaged over eight equally weighted true-orientation bins (rows marked *balanced*).

control	substrate	error
standard decoder (3 seeds)	twin	20.2°
row statistics only, no relational term	twin	37.7°
Gram row only, no row statistics	twin / in vivo	20.3° / 26.5°
row statistics + Gram as attention bias (3 seeds)	twin	$19.9^\circ \pm 0.3^\circ$
disjoint stimulus bins, training vs test (3 seeds)	twin	$20.7^\circ \pm 0.4^\circ$
standard decoder (3 seeds)	in vivo	25.4°
disjoint stimulus bins (60 per side; 3 seeds)	in vivo	$33.8^\circ \pm 0.3^\circ$
same 60 bins on both sides	in vivo	25.3°
neuron split 1 / 2	in vivo	25.4° / 24.6°
standard decoder, raw / balanced	twin	20.3° / 24.5°
orientation-balanced loss, raw / balanced	twin	21.4° / 25.3°
standard decoder, raw / balanced	in vivo	24.6° / 27.9°
orientation-balanced loss, raw / balanced	in vivo	25.8° / 29.1°
class-balanced populations (64 per class, 2 seeds), raw / aligned	twin	22.0° / 21.9°
class-balanced populations, raw / aligned, 2 seeds	in vivo	26.6° / 26.4°
trained on V1, scored on V1 (128), 3 seeds	twin	$22.4^\circ \pm 0.5^\circ$
trained on V1, scored on RL (512), raw / aligned, 3 seeds	twin	26, 43, 46 / 25, 41, 39
trained on V1, scored on AL (128), raw / aligned, 3 seeds	twin	58, 33, 32 / 32, 32, 32
same, class-balanced training (2 seeds), V1 / RL / AL, raw	twin	31° / 29° / 45°
aligned	twin	31° / 29° / 36°
within-scan populations of 128, 3 seeds	twin / in vivo	$32.0 \pm 1.1^\circ$ / $31.1 \pm 1.2^\circ$
one training scan (140 neurons), populations of 128	twin / in vivo	38° – 44° / 37° – 39°
mixed-scan populations of 128, 3 seeds	twin / in vivo	$22.3 \pm 0.4^\circ$ / $25.8 \pm 0.2^\circ$
Allen, mixed populations of 128, pool 280 / 560 / 6,706 cells	2.2M	40.5° / 40.7° / 36.0°

A subtraction ablation that leaks labels. To remove the class-level anisotropy from the input, we estimated the class-Gram residual (8 or 24 classes) on the labelled training half and subtracted the class-pair offset from every Gram entry, using each neuron’s true class, at training and test. The twin error was unchanged (20.7° raw and aligned at 8 classes) and improved at 24 classes (17.7°); the control that subtracts the whole class structure, which should destroy decoding, improved it further (17.2° raw, 15.6° aligned). An offset keyed to the true class is a label written into the Gram, through its jumps at class boundaries and its estimation noise, and the decoder reads it. No result of this ablation is used.

Synthetic populations. 6,000 model neurons with von Mises tuning ($\kappa = 2$), preferred orientations either uniform or with half the neurons within about 12° of a cardinal axis (count), tuning either isotropic or sharpened to $\kappa = 4$ at the cardinals (sharp), and a 120-frame stimulus that either covers orientation evenly or has half its frames near a cardinal axis (stim); responses with Gaussian noise; the 2.2M decoder on 512-neuron populations, epoch selected on the raw error as in every other run, five data and decoder seeds (Table 6). With every switch off the class-Gram is circulant to within noise (neighbour-correlation range 0.01 against 0.04 to 0.14 with a switch on), and the decoder recovers the ordering but no frame. Sharper cardinal tuning and a cardinal excess of neurons fix the frame in at most one seed of five, alone, together, or at any strength (κ up to 32; gain up to 5 times; two seeds each, all above 42° raw); stimulus anisotropy fixes it in most seeds. The anisotropy the cortical data show (Section 3.2) is modelled as a shared stimulus-driven component among like-tuned neurons: a random component per frame on 36 orientation nodes, smoothed over the circle with a von Mises kernel ($\kappa = 4$, half-width about 30°) so that only neurons with nearby preferences share it, enters each neuron’s response with a weight peaked at 90° , with uniform counts, isotropic tuning and an even stimulus. Its class-Gram residual has the one-peaked form of the cortical one and correlates 0.39 to 0.48 with the in-vivo V1 residual; its size is set by the weight and reported as the residual’s fraction of class-Gram variance (V1: 0.16 on the twin, 0.30 in vivo) together with the anisotropy factor (1.67, 1.99 and 1.92 at the three sizes). A global field over orientation with three Fourier modes also fixes the frame (four of five seeds at factor 1.32 and residual fraction 0.01, five of five at 1.68 and 0.06 and at 1.75 and 0.07), but its residual correlates negatively with the cortical one (-0.38), so the local field is the model reported. The spread across seeds is large in every cell where the frame is not fixed, because selection on the raw error then has no signal to select on, and the cell means are therefore reported with their range. The frame-fixed counts are unchanged with the raw-error threshold at 25° or 35° , except that the count cell drops to 0 of 5 at 25° . One run (all three switches, seed 0) never left its initial loss and recovered neither ordering nor frame; it is included in the mean.

Table 6: Synthetic populations, five seeds: raw error / error after the best global rotation or reflection, mean over seeds with the range in brackets, and the number of seeds in which the frame was fixed (raw error below 30° and within 5° of the aligned error). Bottom block: a shared stimulus-driven component among like-tuned neurons (a local random field over preferred orientation, weight peaked at 90°), at three sizes of the class-Gram residual it produces (V1: 0.16 twin, 0.30 in vivo). A gap between raw and aligned error means structure without frame.

anisotropy switched on	raw	aligned	frame fixed
none (circulant)	53.7° [44–63]	30.0° [23–42]	0 / 5
stimulus	23.5° [4–54]	13.9° [4–19]	3 / 5
sharper cardinal tuning	51.4° [39–65]	28.6° [15–42]	0 / 5
cardinal excess of neurons (count)	37.8° [27–45]	35.1° [27–42]	1 / 5
sharper tuning + stimulus	12.2° [3–38]	6.3° [3–10]	4 / 5
count + stimulus	25.0° [4–47]	20.7° [4–43]	3 / 5
count + sharper tuning	42.8° [29–58]	35.2° [27–41]	1 / 5
all three	18.4° [3–46]	17.9° [3–44]	4 / 5
shared component among like-tuned neurons, residual 0.05	3.3° [3–4]	3.2° [3–4]	5 / 5
residual 0.15 (twin V1: 0.16)	2.4° [2–3]	2.4° [2–3]	5 / 5
residual 0.23 (in vivo V1: 0.30)	2.1° [2–2]	2.0° [2–2]	5 / 5

C Receptive-field decomposition per split

Per held-out-mouse split, for the two Allen regimes of Fig. 6: R^2 on absolute screen coordinates, the correlation between predicted and true per-mouse mean position over the 9 test mice (x, y),

and the correlation between predicted and true position relative to the mouse mean over the cells (x, y) .

Table 7: Allen receptive fields per split.

populations	split	absolute R^2	per-mouse mean $r(x, y)$	within-mouse $r(x, y)$
single-animal	0	0.29	0.47, 0.74	0.07, 0.11
single-animal	1	-0.02	0.48, 0.40	0.06, 0.08
single-animal	2	-0.45	0.19, 0.25	0.11, -0.01
mixed	0	0.22	0.70, 0.81	0.00, 0.20
mixed	1	0.08	0.23, 0.80	0.02, 0.08
mixed	2	-0.25	0.46, 0.75	-0.06, 0.04

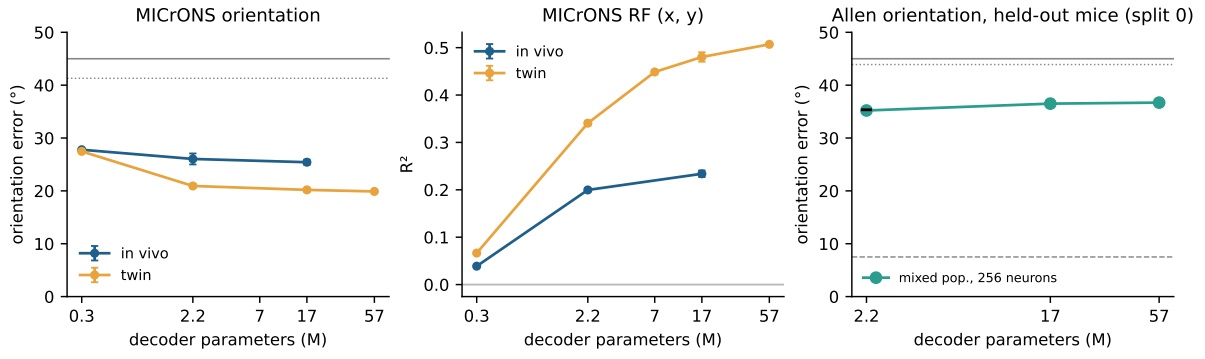


Figure A1: Scaling with decoder size. Test metric versus number of decoder parameters at fixed population size (512 neurons MICrONS, 256 Allen). Error bars: s.d. over 3 seeds where run. On MICrONS receptive fields the gain continues to 57M; the orientation error is flat from 2.2M upward on both substrates and on Allen.

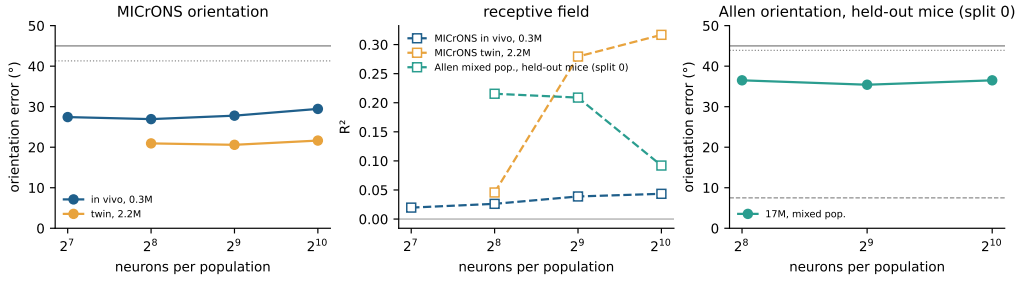


Figure A2: Population size. Test metric versus the number of neurons in each sampled population. Beyond 256 to 512 neurons the orientation error moves by at most 2°, up or down; twin receptive fields keep gaining slowly, Allen receptive fields lose.

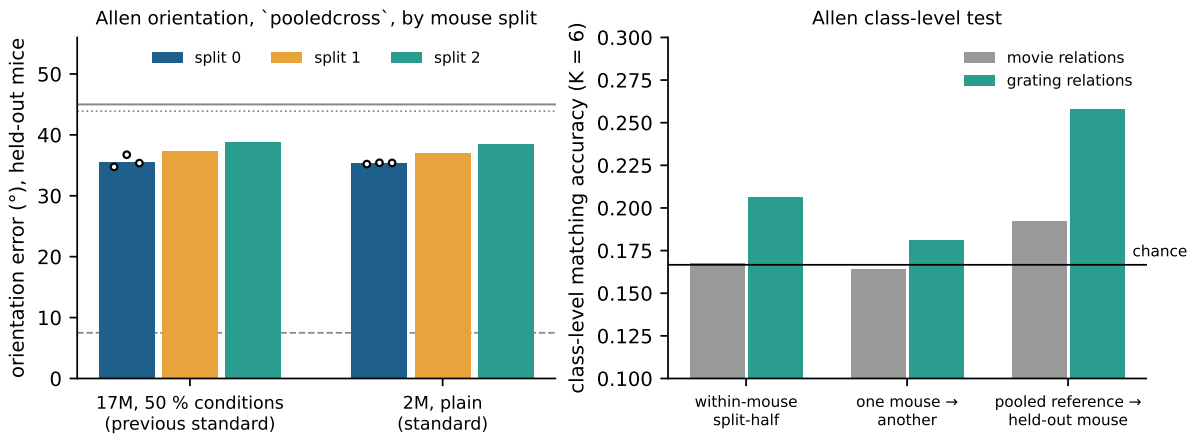


Figure A3: Allen split variance and the class-level test. Left: `pooledcross` orientation error on the same three held-out-mouse splits for the previous standard configuration (17M, augmented) and the final one (2.2M, plain); split 0 carries three seeds. Right: the class-level test with relations from natural movies versus drifting gratings; only the grating relations move every test off chance.